

Metaheuristic Hyperparameter Optimization for Intraday Financial Trend Classification: A Benchmark Across Classifier Backbones

Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound $0.60 \text{ MCC} + 0.40 F_1$ fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

Keywords: Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

1 Introduction

Short-term price forecasting in high-frequency derivatives markets represents one of the most computationally demanding applications of neural computing [1, 2]. Among traded financial instruments, futures contracts are particularly sought after by institutional and retail market participants due to their high liquidity, leverage potential, and utility for both risk hedging against price fluctuations and speculative trading

strategies [3, 4]. These contracts establish an agreement to purchase or sell an underlying asset at a predetermined price on a future date, rendering them exceptionally sensitive to order flow dynamics, short-term market expectations, and macroeconomic announcements [5, 6]. In Brazil, the Mini-Index futures contract (symbol: WIN) on Brasil Bolsa Balcão (B3), referenced to the Ibovespa stock index, exhibits immense daily trading volume and constitutes a prime domain for computational intelligence research in financial time series [1, 7]. However, price series derived from 5-minute intraday candles exhibit severe non-stationarity, low signal-to-noise ratios, structural regime shifts, and rapid dissipation of arbitrage patterns as market participants exploit temporary inefficiencies [8, 9]. These characteristics render accurate directional trend classification a methodologically challenging task that demands both robust feature engineering and principled hyperparameter optimization [10].

Early attempts to formalize market patterns and predict directional movements relied on static linear models, such as autoregressive integrated moving average (ARIMA) techniques [11]. As finance and computer science converged under the computational intelligence paradigm [12], data-driven machine learning (ML) models gained widespread adoption for financial classification tasks. Supervised learning algorithms—including Random Forests (RF) [13], Support Vector Machines (SVM) [14], Multilayer Perceptrons (MLP) [15], and 1D Convolutional Neural Networks (1D-CNN) [16]—have demonstrated strong capabilities in learning complex, non-linear decision boundaries from historical market features. A central prerequisite for building effective classifiers involves technical analysis, wherein mathematical transformations of price and volume series—such as Simple Moving Averages (SMA), Exponential Moving Averages (EMA), and standard deviation envelopes—are computed to capture trend momentum and volatility regimes [10, 17]. Nevertheless, computing high-dimensional pools of technical indicators introduces significant redundancy and noise, creating a feature space that degrades classifier generalization and increases computational overhead [18, 19]. Consequently, applying feature selection techniques to extract compact, highly informative indicator subsets becomes vital for stabilizing training dynamics and improving out-of-sample robustness [3, 20].

Beyond feature engineering and selection, classifier performance depends critically on hyperparameter configuration [21]. In neural network architectures such as MLPs and 1D-CNNs, structural parameters (e.g., hidden layer width, filter counts), regularization coefficients (L2 weight decay α , dropout rate p_{drop}), and training settings (learning rate lr , mini-batch size B) jointly define a complex, non-convex optimization surface over a mixed discrete-continuous search space $\theta \in \mathbb{R}^D$ [9, 21]. Traditional gradient-based algorithms such as Stochastic Gradient Descent or Adam operate effectively on differentiable weight-loss surfaces but cannot directly navigate the non-differentiable, combinatorial hyperparameter landscape [22, 23]. To overcome these limitations, metaheuristic optimization algorithms (MOAs)—spanning evolutionary algorithms such as Genetic Algorithms (GA) [24] and Differential Evolution (DE) [25], as well as swarm intelligence paradigms such as Particle Swarm Optimization (PSO) [26] and Grey Wolf Optimizer (GWO) [22]—have gained substantial prominence for automated hyperparameter tuning in neural computing applications [1, 4]. These population-based methods maintain diverse candidate pools and employ stochastic

exploration-exploitation operators, enabling effective traversal of rugged, multi-modal fitness landscapes without requiring gradient information [22].

Despite the growing literature on metaheuristic-tuned financial classifiers published in premier neural computing journals [1, 4, 9, 16], existing financial machine learning studies overwhelmingly suffer from three major methodological flaws:

1. **The Budget Inequality Flaw:** Benchmark studies frequently evaluate metaheuristics under unconstrained or unequal evaluation counts (e.g., comparing a GA running for 100 generations with a PSO running for 30 iterations) [9]. Because stochastic search performance scales monotonically with candidate evaluation count, reported performance superiority in prior studies is often an artifact of unequal evaluation budgets rather than inherent algorithmic efficiency.
2. **The Temporal Data Leakage Flaw:** Standard randomized K -fold cross-validation or randomized data shuffling is routinely applied to time series data [8, 9]. Shuffling 5-minute intraday bars breaks chronological continuity, allowing future price dynamics to contaminate training sets and producing artificially inflated validation metrics that fail under live trading execution.
3. **The Classification-Economic Disconnect Flaw:** Models are overwhelmingly tuned and evaluated solely on raw classification accuracy or cross-entropy loss [27]. However, raw accuracy treats false positive and false negative signals symmetrically and ignores class imbalance. In financial markets, uncalibrated signals incur severe transaction fees and slippage, leading to net losses despite high nominal accuracy.

To resolve these methodological limitations, this paper presents a controlled multi-optimizer benchmark framework for short-term neural intraday trend classification on 5-minute B3 Mini-Index futures data. Incorporating Information Gain feature selection to extract a compact 7-indicator subset from 66 raw technical indicators, we systematically evaluate five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—across four diverse classifier backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly equal candidate evaluation budget cap of $N_{\text{eval}} = 1,000$ evaluations per seed within a leakage-free 60/20/20 chronological split, contrasting a compound MCC-driven objective ($0.60\text{MCC} + 0.40F_1$) against weighted accuracy fitness. Through non-parametric Friedman and post-hoc Wilcoxon–Holm analyses, we examine optimizer performance across matched blocks and validate configurations via out-of-sample financial backtesting under transaction costs.

The remainder of this manuscript is organized as follows: Section 2 presents a systematic review of the literature, establishing current state-of-the-art paradigms and critical research gaps. Section 3 details the dataset, Information Gain feature selection, model backbones, search-space formulations, metaheuristic operators, and evaluation protocol. Section 4 presents the empirical benchmark results, convergence dynamics, statistical analyses, and financial backtest outcomes. Section 5 interprets search dynamics, trade-offs, and practical deployment implications. Finally, Section 6 concludes the paper and outlines future research directions.

2 Systematic Review of Literature

The intersection of computational intelligence, metaheuristic optimization, and financial time-series forecasting has generated an extensive body of literature over the past decade [12, 23, 28]. Comprehensive surveys have categorized machine learning applications in financial prediction [12, 28], systematically reviewed metaheuristic search paradigms such as Particle Swarm Optimization [29, 30] and Grey Wolf Optimizer [22], and benchmarked neural forecasting architectures under computational constraints [9, 21]. However, despite substantial published progress in premier neural computing journals [1, 4, 10, 16], existing reviews and empirical benchmarks have not systematically evaluated the interaction between equal-budget metaheuristic tuning, temporal validation integrity, and out-of-sample trading performance. This section presents a Systematic Review of Literature (SRL) guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol, examining prior work across feature selection, model backbones, metaheuristic search, and validation protocols, synthesizing current state-of-the-art contributions and formalizing the three critical methodological gaps that motivate our benchmark framework.

2.1 Literature Selection Protocol and Research Questions

To ensure methodological rigor and eliminate reporting bias, relevant literature was retrieved following an explicit PRISMA search protocol across major scholarly databases: IEEE Xplore, Scopus, Web of Science, and SpringerLink (specifically targeting *Neural Computing and Applications*). The search was executed using the following structured Boolean query:

- ("metaheuristic" OR "genetic algorithm" OR "particle swarm" OR "grey wolf")
- AND ("feature selection" OR "technical indicators" OR "Information Gain")
- AND ("futures market" OR "intraday forecasting" OR "trend classification")

The review was bounded to peer-reviewed journal articles and top-tier conference proceedings published between 2019 and 2026. Candidate studies were screened against explicit eligibility criteria:

- **Inclusion Criteria:** (i) evaluation of automated feature selection or metaheuristic hyperparameter tuning on financial or complex time-series classification/forecasting; (ii) explicit disclosure of validation protocols and search space boundaries; and (iii) reporting of empirical predictive or out-of-sample financial metrics.
- **Exclusion Criteria:** Studies lacking reproducible experimental parameters, omitting candidate evaluation budget details, or failing to disclose optimization constraints were excluded.

Figure 1 depicts the complete PRISMA literature selection workflow. The systematic review is organized around three central Research Questions (RQs):

1. **RQ1 (Budget Parity)**: How do stochastic metaheuristic search operators (RS, GA, PSO, DE, GWO) compare across distinct classifier backbones (RF, SVM, MLP, 1D-CNN) when restricted to a strictly equal candidate evaluation budget cap ($N_{\text{eval}} = 1,000$)?
2. **RQ2 (Temporal Leakage)**: What is the quantitative impact on out-of-sample generalization when comparing chronological sequential temporal splits against randomized cross-validation on high-frequency financial time series?
3. **RQ3 (Economic Alignment)**: How effectively do classification-based validation fitness objectives (e.g., compound MCC/ F_1 vs. weighted accuracy) align with out-of-sample intraday trading metrics (Net Return %, Sharpe ratio, Maximum Drawdown) under realistic exchange transaction costs?

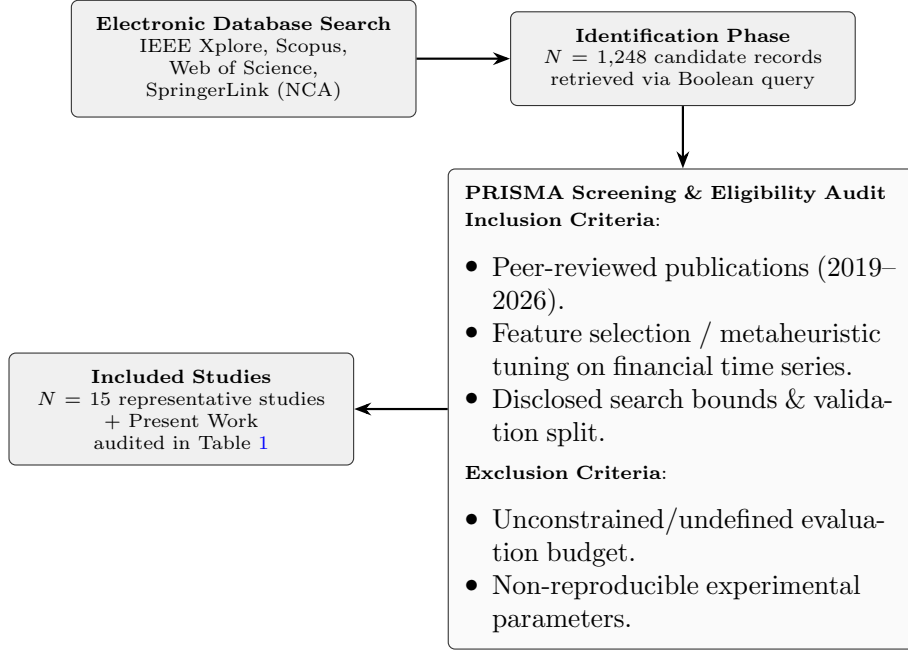


Fig. 1 PRISMA literature selection protocol flowchart mapping database retrieval, eligibility criteria filtering, and final synthesis of 15 representative studies audited in Table 1.

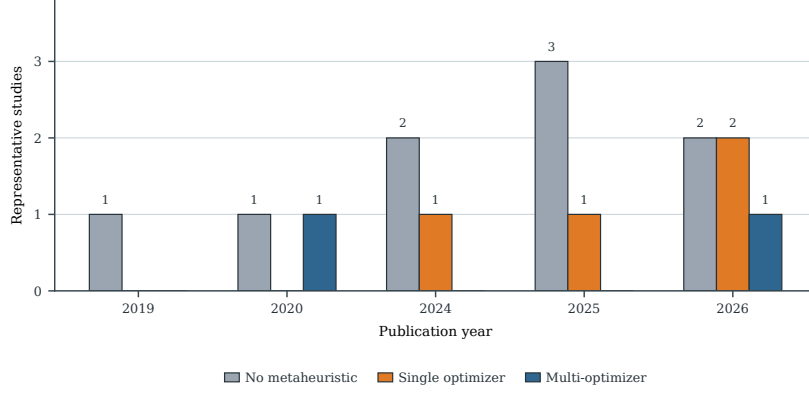


Fig. 2 Distribution of the 15 representative prior studies included in the literature synthesis, grouped by publication year and the scope of optimizer comparison. A study is assigned to one category according to its primary experimental design; the present work is excluded. The limited number of multi-optimizer studies motivates the controlled benchmark developed in this paper.

2.2 Classifier Backbones and Input Representations

2.2.1 Technical indicators as input representations

The quality and information density of the input representation are central determinants of supervised intraday trend prediction [18]. Open, high, low, close, and volume observations encode the market state, but short-horizon bars are also affected by microstructural noise, regime shifts, and high variance. For this reason, financial machine-learning studies commonly transform raw bar data into technical descriptors representing complementary aspects of price behaviour: momentum, trend, dispersion, and volatility [10, 17, 31]. Such representations do not create information; rather, they provide a structured hypothesis space through which a learner can test whether recent price and volume dynamics carry directional content.

The expansion of the indicator set must nevertheless be controlled. Indicators derived from common prices and overlapping lookback windows are often strongly correlated, so a large technical library can amplify redundancy, increase estimation variance, and make model ranking sensitive to incidental patterns in the development sample [19, 20]. Feature selection is therefore a component of model design rather than a preliminary convenience: it reduces the dimensionality exposed to subsequent learning and hyperparameter search, while retaining descriptors with a demonstrable association with the target. Information Gain (IG) is a transparent filter criterion for this purpose, ranking candidate indicators by their reduction of class uncertainty [3, 19]. Although a univariate ranking does not guarantee an optimal multivariate subset, its low computational cost and auditability make it appropriate for constructing a compact technical representation. The mathematical formulation and empirical application of IG in the present benchmark are detailed in Section 3.3.

2.2.2 Classifier families

The literature spans several classifier families whose distinct inductive biases make a single preferred backbone unlikely across all market representations:

- **Ensemble decision trees**, represented by Random Forest (RF), aggregate decorrelated trees built from bootstrap samples and randomly selected feature subsets. Their partition-based structure accommodates non-linear interactions and can be relatively robust to noisy descriptors [13].
- **Kernel margin classifiers**, represented by Support Vector Machines (SVM), seek separating hyperplanes with a maximum-margin principle. Kernel mappings, including the radial-basis-function kernel, permit non-linear decision surfaces while retaining a well-defined optimization objective [14].
- **Feedforward neural networks**, represented by Multilayer Perceptrons (MLP), learn non-linear mappings through stacked fully connected layers and backpropagation. Their expressive capacity is accompanied by sensitivity to architectural and regularization choices [15].
- **One-dimensional convolutional neural networks** (1D-CNN) apply shared local filters to ordered input windows. This construction can capture short-range patterns while using parameter sharing to constrain the number of learned weights [9, 16].

These families therefore represent different hypotheses about how directional information is encoded. Their comparative use is more informative than selecting a single algorithm a priori, particularly because changes in feature representation and hyperparameterization can alter the observed ranking of classifiers [13, 32]. The specific representations, search domains, and fitted architectures adopted in this study are defined in Section 3.

2.3 Metaheuristic Optimization Paradigms

2.3.1 Population-based search for hyperparameter selection

Hyperparameter selection is naturally formulated as a black-box optimization problem. Let $\theta \in \Omega$ denote a feasible model configuration, where Ω can contain continuous, integer, categorical, and conditional variables. A validation criterion $f(\theta)$ is obtained only after fitting and evaluating the corresponding learning pipeline, and the search objective is

$$\theta^* = \arg \max_{\theta \in \Omega} f(\theta). \quad (1)$$

Unlike weight training, this objective need not be differentiable with respect to θ . Tree depth, number of hidden units, kernel choices, and regularization settings may also be discrete or conditionally active. Population-based metaheuristics are consequently attractive because they navigate such mixed spaces without gradient information and can balance broad exploration against local refinement [21–23].

In this context, a “candidate evaluation” should mean one complete configuration assessment under an unchanged validation protocol. This distinction is important: iterations and generations are internal counters, whereas the number of evaluated

candidates is the common computational unit through which heterogeneous search procedures can be compared.

Figure 3 provides a compact taxonomy of the learning and search families considered in this literature. It distinguishes the classifier hypothesis spaces from the mechanisms used to select their configurations, while retaining the shared objective of validation-based model selection.

2.3.2 Evolutionary paradigms

Evolutionary algorithms maintain a population of candidate configurations and improve it through selection and variation. Genetic Algorithms (GA) emulate selection, crossover, and mutation: configurations with stronger fitness are preferentially retained or recombined, while mutation introduces new material into the population [21, 24]. This representation-agnostic cycle is well suited to mixed hyperparameter spaces after appropriate encoding and constraint handling.

Differential Evolution (DE) is also population based, but creates candidate displacements from scaled differences between population members. A canonical donor-vector form is

$$\mathbf{v}_i^t = \mathbf{x}_{r_1}^t + F(\mathbf{x}_{r_2}^t - \mathbf{x}_{r_3}^t), \quad (2)$$

where r_1 , r_2 , and r_3 index distinct population members and F scales the differential displacement. Crossover then combines the donor with a target vector, and selection retains the better candidate [25]. The use of population differences gives DE a self-adaptive search scale, although its practical behaviour still depends on encoding, bounds, and the chosen mutation and crossover strategy.

2.3.3 Swarm-intelligence paradigms

Swarm methods update candidates by combining individual experience with population-level information. In Particle Swarm Optimization (PSO), each particle has a position $\mathbf{x}_i^t$, a velocity $\mathbf{v}_i^t$, a personal best position $\mathbf{p}_i^t$, and an incumbent swarm-best position $\mathbf{g}^t$. Its standard update is

$$\mathbf{v}_i^{t+1} = \omega \mathbf{v}_i^t + c_1 r_1 (\mathbf{p}_i^t - \mathbf{x}_i^t) + c_2 r_2 (\mathbf{g}^t - \mathbf{x}_i^t), \quad (3)$$

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}, \quad (4)$$

where ω controls inertia, c_1 and c_2 weight cognitive and social attraction, and r_1, r_2 are random variates. This formulation makes the exploration–exploitation trade-off explicit and is the foundation for the broad family of PSO variants reviewed by Gad [26, 29].

Grey Wolf Optimizer (GWO) uses a different collective metaphor. Candidate solutions are ranked into leading α , β , and δ wolves, while the remaining wolves update their positions by encircling and approaching these leaders. Its control schedule progressively shifts the search from exploration to exploitation [4, 22]. Despite distinct metaphors, PSO and GWO share the same benchmarking requirement: each update must be translated into feasible configurations and assessed under the same validation rule.

2.3.4 Baselines and implications for comparative studies

Random Search (RS) is an indispensable baseline because it samples feasible configurations without an adaptive search trajectory. It establishes whether the additional mechanisms of evolutionary or swarm search improve solution quality beyond unguided sampling from the same search space [33]. A credible comparison must therefore control the parameter domain, candidate-evaluation count, validation data, random replication, and reporting metrics. Equalizing only the nominal number of generations or iterations is insufficient when algorithms evaluate different numbers of candidates per update.

The comparative evidence also highlights recurring weaknesses: unequal computational budgets can confound search quality with search effort; randomized validation can compromise temporal integrity; and classification-only reporting may not establish economic value after transaction costs [10, 15, 16]. These issues are synthesized in Section 2.5; the concrete controls used by the present benchmark are specified only in Section 3.

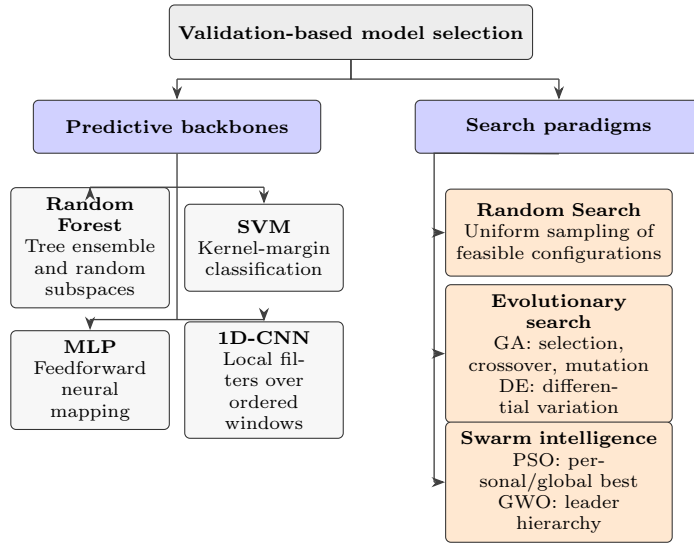


Fig. 3 Conceptual taxonomy linking predictive backbones with hyperparameter-search strategies in the reviewed literature. The left branch denotes alternative classifier hypothesis spaces; the right branch distinguishes unguided sampling, evolutionary variation, and swarm-based collective search. The figure is conceptual; the precise search domains and experimental controls are specified in Section 3.

2.4 Comparative Synthesis of Representative Studies

To establish the state of the art and pinpoint persistent methodological gaps, Table 1 synthesizes 15 representative published studies from 2019 to 2026 alongside the present benchmark work. Studies are evaluated across six technical dimensions: Study & Year,

Application Domain & Data, Model Backbones, Optimization & Feature Paradigms, Validation Protocol, and Identified Methodological Gap.

Table 1 Systematic Literature Review comparison matrix auditing representative studies in financial machine learning, metaheuristic hyperparameter tuning, and temporal validation protocols against identified methodological gaps.

Study & Year	Domain & Data	Models	Optimization & Features	Validation Protocol	Methodological Gap
Nabipour et al. (2020) [13]	Stock index trend	RF, SVM, ANN, XGBoost	Technical indicators	Train/test split	Absence of metaheuristic hyperparameter tuning.
Jovanovic et al. (2024) [21]	Crop-yield prediction	Weight-agnostic neural networks	Metaheuristic tuning	Train/test evaluation	Non-financial domain; no equal-budget financial benchmark.
Naik & Mohan (2019) [14]	Equities (NSE India)	ANN classifier	Boruta feature selection	Train/test split	Static parameters; no metaheuristics.
Ecer et al. (2020) [15]	Financial trend	MLP classifier	GA and PSO tuning	K -fold CV	Unconstrained budget; single classifier.
Billah et al. (2024) [10]	Intraday equities	Heuristic models	Moving average rules	Standard split	Absence of neural hyperparameter tuning.
Bareket & Párv (2024) [34]	Surge detection	Adaptive ML	Technical indicators	Medium-term split	Evaluation restricted to static features.
Bandgar et al. (2025) [16]	Financial forecasting	Ensembles	Default tuning	Random split	No evaluation budget cap; data leakage.
Rahim et al. (2025) [35]	5-min futures	Single neural model	Single GA optimization	Train/test split	Single GA; no multi-swarm benchmark.
Dhingra et al. (2025) [9]	Solar PV forecasting	LSTM, CNN-LSTM	RAdam, Lookahead, Adam	Sequential split	Non-financial domain; stochastic optimizers omitted.
Mutinda & Yong (2026) [32]	Financial market	Ensembles	SEDT feature selection	K -fold CV	Feature selection without tuning.
Simatupang et al. (2026) [36]	Stock index (IDX)	Bayesian RF	Bayesian optimization	Random K -fold CV	Random CV causes temporal data leakage.
Al Bannoud et al. (2026) [4]	Kinetics time series	ANN-ODE surrogate	GWO, PSO, GA, ABC	Multi-partition	Non-financial; unconstrained search budget.
Bashir et al. (2026) [1]	Equities (NSE India)	Bagging Ensemble	(1+1) Evolutionary Algo	Train/test split	Single EA evaluated without budget cap.
[Anonymized Precursor A] (2025) [3]	Intraday B3 futures	Random Forest	InfoGain + Default RF	Sequential 60/20/20	Untuned baseline classifier without metaheuristics.
[Anonymized Precursor B] (2026) [37]	Intraday B3 futures	MLP, SVM, RF	InfoGain + Single GA	5-fold CV	Single GA; unconstrained 1,000-generation budget.
Present Work	Intraday B3 futures	MLP, SVM, RF, 1D-CNN	InfoGain + RS, GA, PSO, DE, GWO	Sequential 60/20/20	Equal budget ($N_{\text{eval}} = 1,000$), 3 seeds, fee backtest.

2.5 Critical Methodological Gaps and Research Motivation

As synthesized in Table 1, three primary methodological flaws permeate the existing literature on metaheuristic financial forecasting:

1. *The Budget Inequality Flaw*: Prior studies routinely compare metaheuristics without enforcing a strict candidate evaluation cap (N_{eval}). Because stochastic optimization performance scales with function evaluations, superiority claims in unconstrained studies often reflect unequal computational budgets rather than algorithmic search efficiency [9, 15].

2. *The Temporal Data Leakage Flaw*: Financial machine learning research frequently relies on randomized K -fold cross-validation or data shuffling [8, 36]. Shuffling sequential intraday price bars allows future market information to contaminate historical training folds, yielding artificially inflated predictive accuracy scores that collapse during live execution.
3. *The Classification-Economic Disconnect Flaw*: Hyperparameter tuning is overwhelmingly conducted against raw accuracy or cross-entropy loss [3, 27]. In high-frequency futures trading, raw accuracy ignores class imbalance and trade execution costs. A classifier achieving high nominal accuracy can incur net financial losses once exchange transaction fees and slippage are deducted.

To address these limitations, this paper establishes an auditable multi-optimizer benchmark framework on 15,057 five-minute B3 Mini-Index futures bars. Leveraging Information Gain feature selection to extract a compact 7-indicator subset, we systematically evaluate five optimizer families (RS, GA, PSO, DE, GWO) across four diverse model backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly locked candidate evaluation cap of $N_{\text{eval}} = 1,000$ evaluations per seed across three common random seeds within a leakage-free 60/20/20 chronological partition. Furthermore, we contrast two distinct validation fitness objectives (a compound $0.60 \text{ MCC} + 0.40 F_1$ objective vs. weighted accuracy) and evaluate held-out trading profitability under realistic exchange transaction costs.

3 Materials and Methods

This section presents the methodology adopted in this study. Figure 4 provides an overview of the flow from data acquisition to final model evaluation. It preserves the practical sequence of the IJCNN precursor—input data, technical indicators, feature selection, optimization, model fitting, and evaluation—while inserting the temporal controls and the two validation objectives required by the expanded benchmark.

3.1 Input Data

The data were obtained from the BovDB resources [6, 7]. The analyzed subset contains 15,057 clean five-minute records of the WIN contract from January to June 2024. WIN is referenced to the Ibovespa index and was selected because its liquidity and frequent intraday movement make it suitable for short-horizon computational prediction. The binary target indicates the next-bar direction: 0 for downtrend and 1 for uptrend.

The records remain in chronological order. The first 9,034 observations are used for training, the subsequent 3,011 observations for validation, and the final 3,012 observations for locked testing. No random shuffling is applied. Figure 5 makes this separation explicit; the test observations are unavailable during both feature and hyperparameter selection.

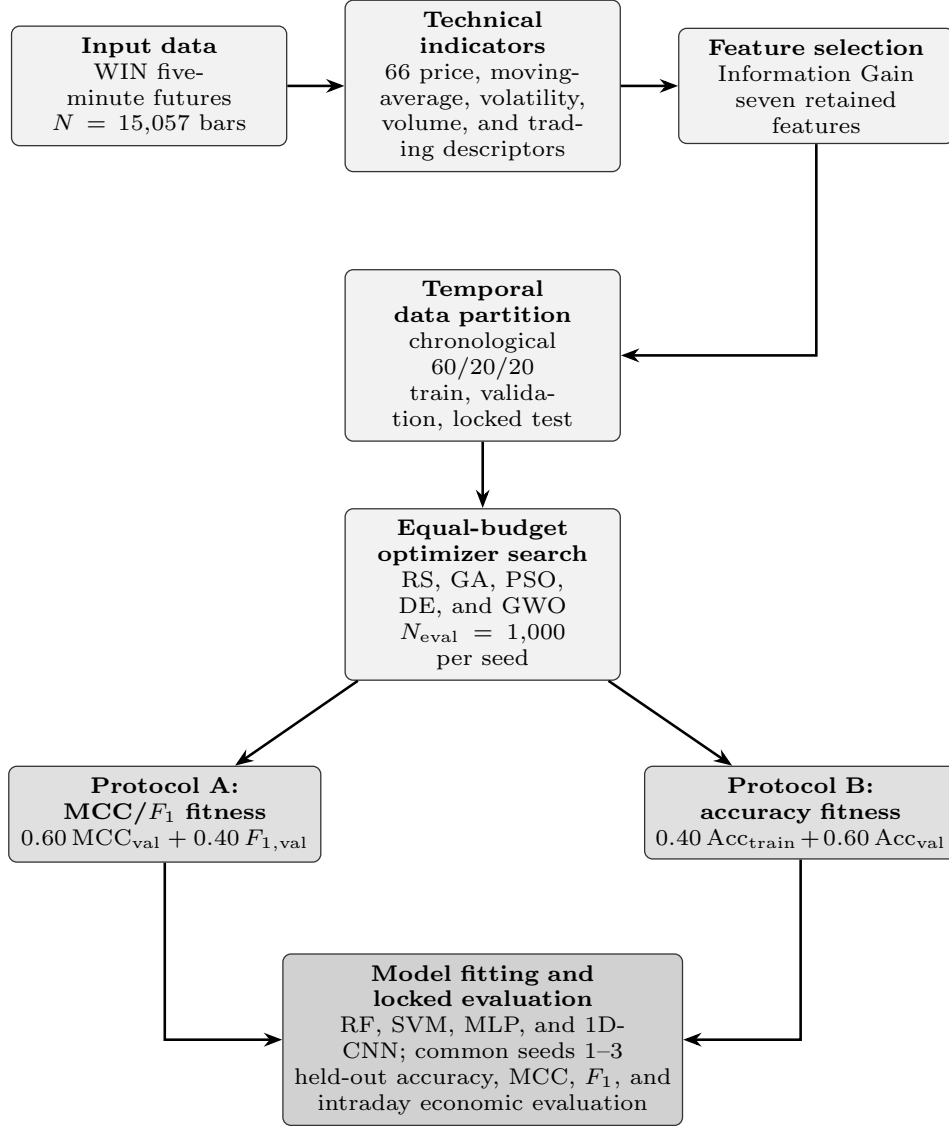


Fig. 4 Overview of the proposed methodology. The new benchmark preserves the precursor workflow while enforcing a chronological split, equal optimizer budget, and two separated validation objectives.

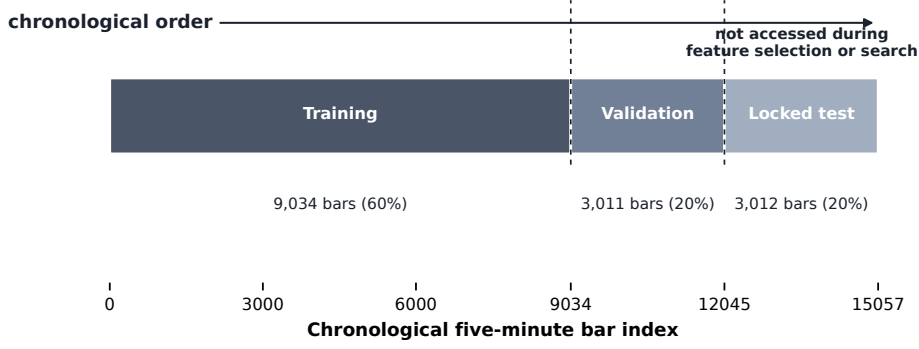


Fig. 5 Chronological data partition used in every experiment. Training, validation, and locked test segments occupy consecutive periods of the WIN five-minute series.

3.2 Generation of Technical Analysis Indicators

Technical indicators were derived from open, high, low, close, trade volume, and transaction-frequency information. Simple Moving Average (SMA), Exponential Moving Average (EMA), and dispersion measures were computed over short windows (3, 5, 7, and 9 periods) to retain sensitivity to intraday changes. For example, the difference between the seven- and three-period SMA is

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3. \quad (5)$$

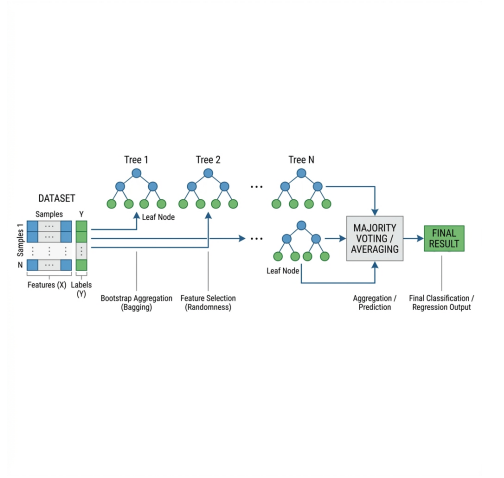
The raw and derived variables were normalized before model fitting. Together, the construction procedure produces 66 descriptors, including moving-average differences and normalized counterparts.

3.3 Feature Selection Method

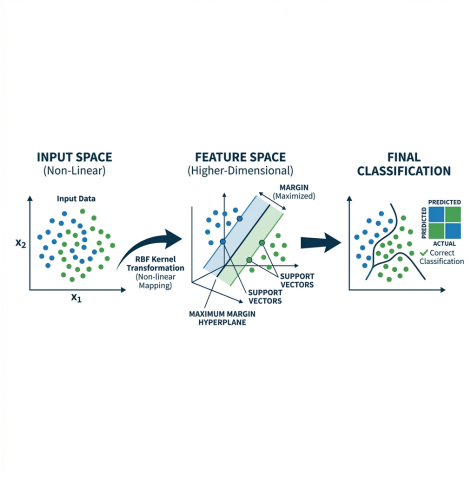
The 66-indicator pool was filtered using Information Gain attribute selection evaluated strictly on the training partition ($N = 9,034$ bars). This information-theoretic ranking isolates attributes that maximize entropy reduction regarding next-bar trend directions while eliminating redundant lookback descriptors [19, 20]. Seven features were retained: EMA_{5-3} , normalized EMA_{5-3} , EMA_{7-3} , normalized EMA_{7-3} , normalized SMA_9 , EMA_{9-3} , and normalized EMA_7 . This identical 7-variable input vector $\mathbf{X}_{\text{selected}} \in \mathbb{R}^7$ is fed into all classifier backbones across all optimizer runs.

3.4 Implemented Classifier Architectures and Search Spaces

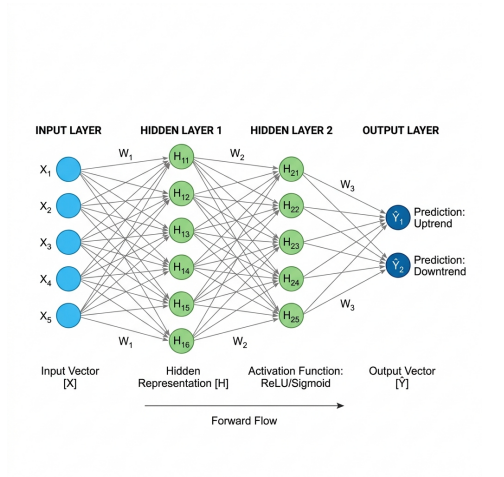
To benchmark metaheuristic search performance across fundamentally distinct hypothesis spaces, four classifier backbones were implemented: Random Forest (RF), Support Vector Machines (SVM), Multilayer Perceptron (MLP), and 1D Convolutional Neural Networks (1D-CNN). Figure 6 illustrates the structural decision mechanisms and layer topologies of these four implemented architectures.



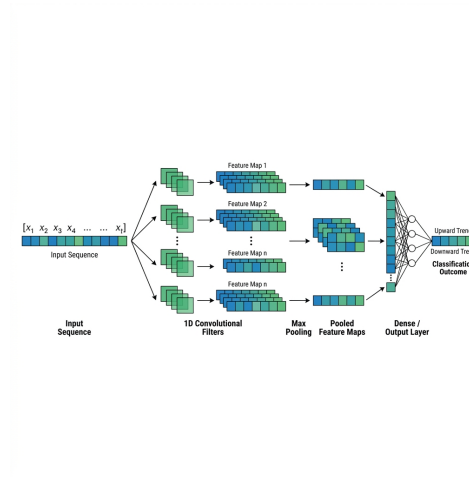
(a) Random Forest



(b) Support Vector Machine



(c) Multilayer Perceptron



(d) 1D Convolutional Neural Network

Fig. 6 Architectural topologies and structural decision mechanisms of the four candidate classifier backbones evaluated under the equal-budget metaheuristic optimization framework.

The hyperparameter search space bounds $\theta \in \mathcal{S}$ for each implemented model are defined as follows:

- **RF**: Number of trees $n_{\text{tree}} \in [10, 200]$, maximum tree depth $d_{\text{max}} \in [2, 20]$, and minimum split samples $s_{\text{min}} \in [2, 10]$.
- **SVM**: Soft-margin penalty $C \in [10^{-2}, 10^2]$ and Radial Basis Function kernel bandwidth $\gamma \in [10^{-3}, 10^1]$ sampled on logarithmic scales.
- **MLP**: Hidden layer width $n_{\text{hidden}} \in [16, 128]$, L2 penalty coefficient $\alpha \in [10^{-4}, 10^{-1}]$, and initial learning rate $\eta_0 \in [10^{-4}, 10^{-1}]$.
- **1D-CNN**: Convolutional filter count $n_{\text{filter}} \in [16, 64]$, kernel size $k_{\text{size}} \in [2, 5]$, dense layer width $n_{\text{dense}} \in [16, 64]$, dropout rate $p_{\text{drop}} \in [0.1, 0.5]$, and Adam learning rate $\eta \in [10^{-4}, 10^{-2}]$.

3.5 Optimizer Configuration, Search Operators, and Validation Objectives

To ensure complete experimental reproducibility, five search paradigms were evaluated: Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO). All optimizers operate under a strict candidate evaluation cap of $N_{\text{eval}} = 1,000$ per model and random seed. For all population-based algorithms (GA, PSO, DE, GWO), the population size is fixed to $N_{\text{pop}} = 10$ candidates evaluated across $G = 100$ generations or iterations ($10 \times 100 = 1,000$ function evaluations). RS generates $N_{\text{eval}} = 1,000$ independent uniform random samples across the search space.

Candidate hyperparameter vectors are encoded as continuous real vectors $\mathbf{u} \in [0, 1]^D$. Discrete parameters (such as tree counts, hidden widths, and filter sizes) are mapped via step quantization:

$$\theta_d = \lfloor \theta_{d,\text{min}} + u_d(\theta_{d,\text{max}} - \theta_{d,\text{min}} + 1) \rfloor. \quad (6)$$

Continuous parameters defined over logarithmic scales (such as regularization parameters C, α and learning rates η) are decoded via exponential scaling:

$$\theta_d = 10^{\log_{10}(\theta_{d,\text{min}}) + u_d(\log_{10}(\theta_{d,\text{max}}) - \log_{10}(\theta_{d,\text{min}}))}. \quad (7)$$

The specific search operators and update formulations governing each optimizer are defined as follows:

- **Genetic Algorithm (GA)**: Utilizes binary tournament selection ($k = 2$), uniform crossover with probability $P_c = 0.8$, and Gaussian mutation with probability $P_m = 0.1$ and standard deviation $\sigma = 0.1$.
- **Particle Swarm Optimization (PSO)**: Velocity vectors $\mathbf{v}_i$ and positions $\mathbf{u}_i$ for particle i at iteration g are updated using inertia weight $w = 0.729$ and cognitive/social coefficients $c_1 = c_2 = 1.494$:

$$\mathbf{v}_i^{g+1} = w\mathbf{v}_i^g + c_1 r_1 (\mathbf{pbest}_i - \mathbf{u}_i^g) + c_2 r_2 (\mathbf{gbest} - \mathbf{u}_i^g), \quad (8)$$

$$\mathbf{u}_i^{g+1} = \text{clip} \left(\mathbf{u}_i^g + \mathbf{v}_i^{g+1}, 0, 1 \right), \quad (9)$$

where $r_1, r_2 \sim U(0, 1)$.

- **Differential Evolution (DE)**: Implements the DE/best/1/bin strategy with mutation factor $F = 0.5$ and binomial crossover probability $CR = 0.7$:

$$\mathbf{v}_i^{g+1} = \mathbf{gbest}^g + F(\mathbf{u}_{r_1}^g - \mathbf{u}_{r_2}^g), \quad (10)$$

where r_1, r_2 are distinct randomly chosen population indices.

- **Grey Wolf Optimizer (GWO)**: Candidate position vectors are updated by simulating encirclement of the three leading wolves (α, β, δ):

$$\mathbf{u}_i^{g+1} = \frac{\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3}{3}, \quad (11)$$

where $\mathbf{x}_1 = \alpha - \mathbf{A}_1 \cdot |\mathbf{C}_1 \cdot \alpha - \mathbf{u}_i^g|$, with analogous terms for $\mathbf{x}_2$ (β) and $\mathbf{x}_3$ (δ).

Candidate configurations are evaluated against two separated validation objectives:

- **Protocol A (MCC/ F_1 Compound Objective)**:

$$f_{\text{MCC}/F_1}(\theta) = 0.60 \text{MCC}_{\text{val}}(\theta) + 0.40 F_{1,\text{val}}(\theta). \quad (12)$$

- **Protocol B (Weighted Accuracy Objective)**:

$$f_{\text{Acc}}(\theta) = 0.40 \text{Accuracy}_{\text{train}}(\theta) + 0.60 \text{Accuracy}_{\text{val}}(\theta). \quad (13)$$

Upon completion of the $N_{\text{eval}} = 1,000$ search budget, the optimal candidate vector $\theta^* = \arg \max f(\theta)$ is refit on the concatenated training and validation dataset ($N = 12,045$ bars) and evaluated exactly once on the locked test block ($N = 3,012$ bars). Out-of-sample directional predictions drive an automated backtest execution engine under an exchange transaction fee of 0.03% per round-trip trade to compute Net Return %, Sharpe ratio, and Maximum Drawdown.

4 Results

This section presents the experimental results and their implications for short-term prediction of Brazilian Mini-Index futures. Following the organization of the precursor, it first describes the common experimental setup and then discusses the performance of the individual model families. The expanded benchmark subsequently reports the direct cross-model comparison, exploratory statistical evidence, and economic outcomes.

4.1 Experimental Setup and Data Partitioning

All model-optimizer combinations use the seven descriptors described in Section 3.3, the chronological split shown in Figure 5, and the same 1,000-evaluation allowance per seed. RF, SVM, MLP, and 1D-CNN each are searched with RS, GA, PSO, DE, and

GWO. The two protocols differ only in the validation fitness used to rank candidate configurations. Three common seeds (1–3) are available for every cell in both protocols.

Consequently, the direct comparison between fitness functions is descriptive: each reported model–optimizer result is the mean over the same three seeds. The locked test block contains 3,012 observations and is not used during search. This design differs from the randomized training and test sampling used in the precursor and removes look-ahead exposure from model selection.

4.2 Performance of the Models

Table 2 retains the presentation pattern of the precursor: it states the search space and an illustrative GA-selected configuration for each of the four model families. The configurations are from seed 1 under Protocol A and are included to make the optimization procedure concrete; they are not asserted to be universally optimal because all optimizer–seed outcomes are summarized below.

Figures 7–10 replace the three GA-only convergence plots of the precursor with the five budget-matched search trajectories. For each model, the left panel follows the compound MCC/ F_1 fitness and the right panel follows weighted training/validation accuracy. The vertical scales differ because the objectives differ; their role is to show search convergence within a protocol, not to compare objective magnitudes across protocols.

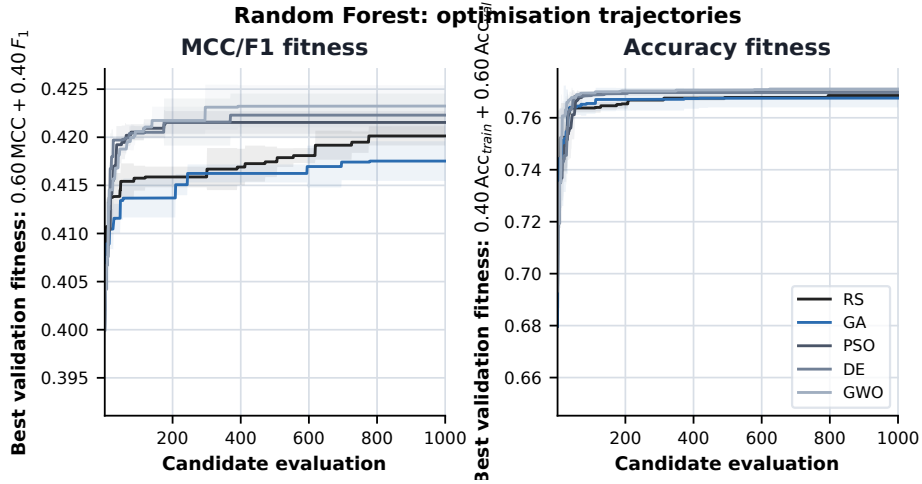


Fig. 7 Fitness evolution during equal-budget optimization of the Random Forest model. Lines summarize the three common seeds.

Table 3 and Figure 11 present held-out performance aggregated over 15 optimizer–seed blocks per model within each protocol. Under MCC/ F_1 fitness, RF has the largest mean test accuracy (0.645), MCC (0.289), and F_1 (0.664). Under weighted accuracy fitness, MLP has the largest mean test accuracy (0.651), MCC (0.301), and F_1 (0.658). The reversal reinforces that the validation criterion and model family should be reported separately rather than treated as interchangeable.

Table 2 Hyperparameter search domains and representative GA-selected configurations across the four evaluated classifier backbones.

Random Forest (RF)			Support Vector Machine (SVM)		
Hyperparameter	Search space	Selected	Hyperparameter	Search space	Selected
$n_{\text{estimators}}$	[50, 500]	58	C	$[10^{-3}, 10^3]$ (log scale)	2.089
max_depth	[5, 50]	6	γ	$[10^{-5}, 10^1]$ (log scale)	5.044
min_samples_split	[2, 20]	13	Kernel	{linear, RBF}	linear
min_samples_leaf	[1, 20]	15			
max_features	{sqrt, log2, None}	None			
class_weight	{None, balanced}	None			
Multilayer Perceptron (MLP)			1D Convolutional Neural Network (1D-CNN)		
Hyperparameter	Search space	Selected	Hyperparameter	Search space	Selected
Hidden neurons	[5, 500]	410	Filters	[8, 128]	48
Learning rate	$[10^{-8}, 10^{-2}]$ (log scale)	3.42×10^{-4}	Kernel size	[2, 5]	3
L_2 regularization	$[10^{-8}, 10^{-2}]$ (log scale)	1.66×10^{-7}	Dense neurons	[16, 256]	251
Dropout rate	[0, 0.10]	0.057	Learning rate	$[10^{-8}, 10^{-2}]$ (log scale)	3.90×10^{-6}
Batch size	{16, 32, 64, 80, 112, 128}	112	L_2 regularization	$[10^{-8}, 10^{-2}]$ (log scale)	5.70×10^{-6}
			Dropout rate	[0, 0.10]	0.026
			Batch size	{16, 32, 64, 80, 112, 128}	16

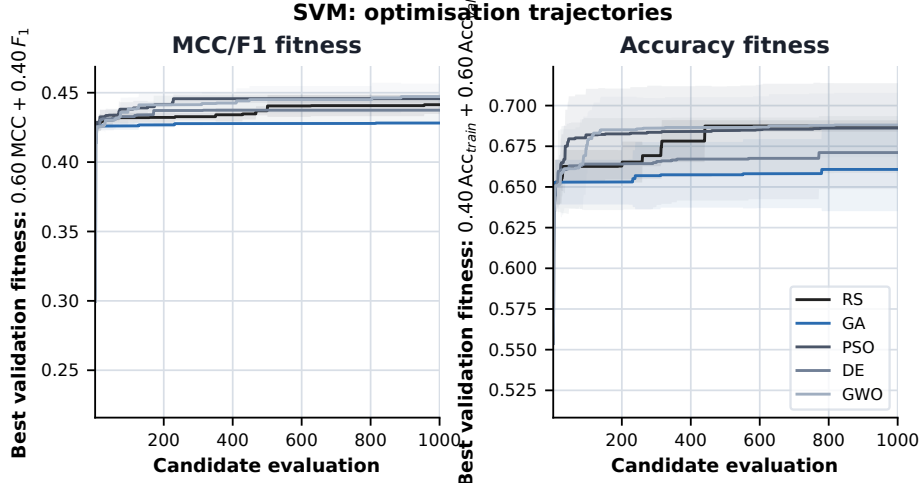


Fig. 8 Fitness evolution during equal-budget optimization of the SVM model. Lines summarize the three common seeds.

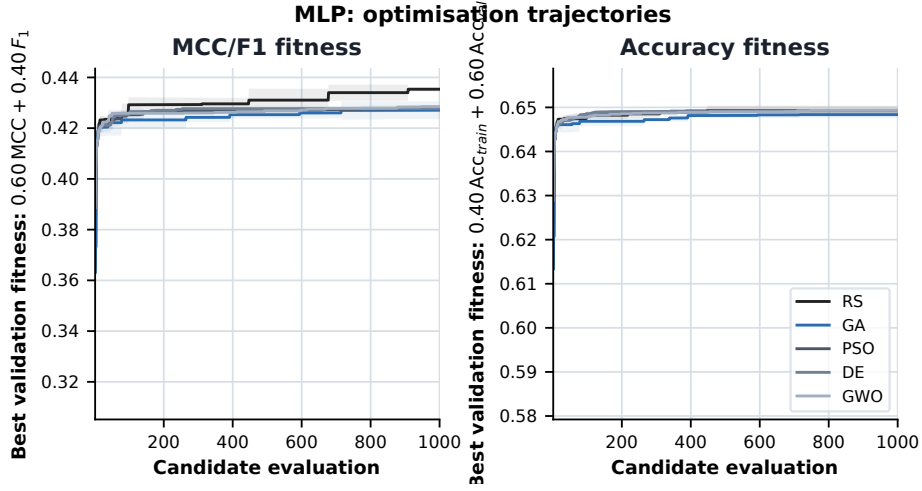


Fig. 9 Fitness evolution during equal-budget optimization of the MLP model. Lines summarize the three common seeds.

Table 3 Mean locked-test predictive outcomes by backbone and validation protocol (15 optimizer-seed blocks per row).

Model	MCC/ F_1 fitness			Weighted accuracy fitness		
	Accuracy	MCC	F_1	Accuracy	MCC	F_1
RF	0.645	0.289	0.664	0.604	0.208	0.624
SVM	0.604	0.214	0.602	0.578	0.159	0.600
MLP	0.630	0.261	0.639	0.651	0.301	0.658
1D-CNN	0.613	0.224	0.597	0.635	0.271	0.639

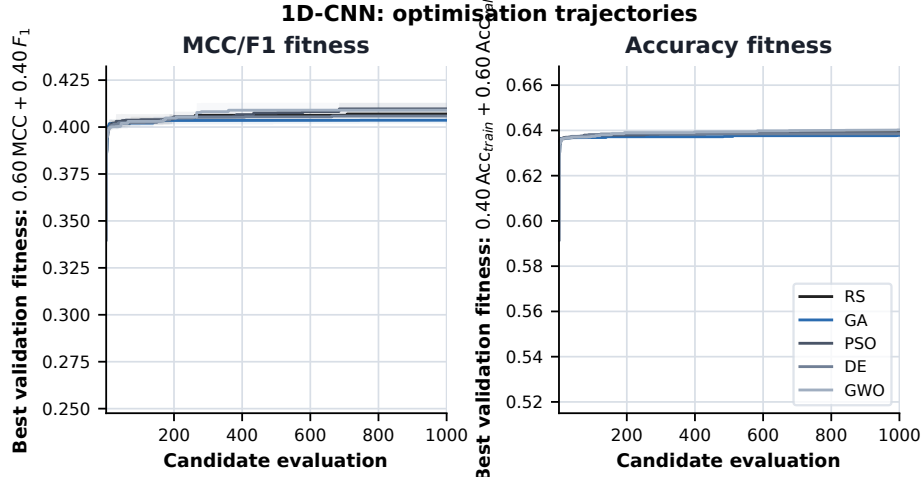


Fig. 10 Fitness evolution during equal-budget optimization of the 1D-CNN model. Lines summarize the three common seeds.

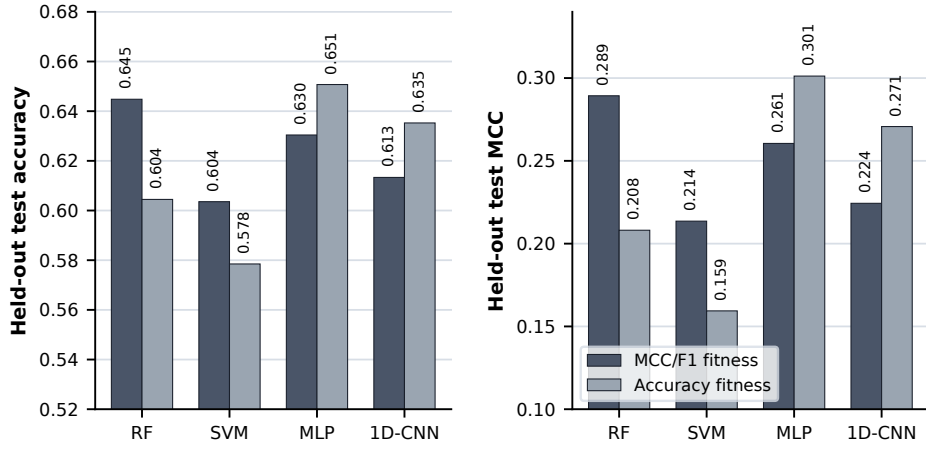


Fig. 11 Comparison of mean locked-test accuracy and MCC for all four model families under the two validation objectives.

4.3 Exploratory Statistical and Economic Evaluation

The model-level tests are performed within each holdout protocol. The five optimizer-seed combinations yield 15 matched blocks for each backbone. Under MCC/ F_1 fitness, the Friedman comparison of test MCC is significant ($\chi^2 = 22.84$, $p = 4.36 \times 10^{-5}$), with RF obtaining the best mean outcome. Under weighted accuracy fitness, the comparison of test accuracy is significant ($\chi^2 = 42.92$, $p = 2.56 \times 10^{-9}$), with MLP first; its Holm-adjusted pairwise contrasts against RF, SVM, and 1D-CNN are significant ($p_{\text{Holm}} \approx 0.00194$). These tests compare models within a protocol and should not be interpreted as a confirmatory test between the two fitness functions.

The economic evaluation yields a complementary ranking. Under MCC/F_1 fitness, RF has the largest mean total profit (26,465 points). Under weighted accuracy fitness, MLP has the largest mean total profit (27,737 points), whereas RF has the least negative mean maximum drawdown (-1,866 points). Figure 12 therefore makes explicit the difference between predictive ranking and return-risk ranking.

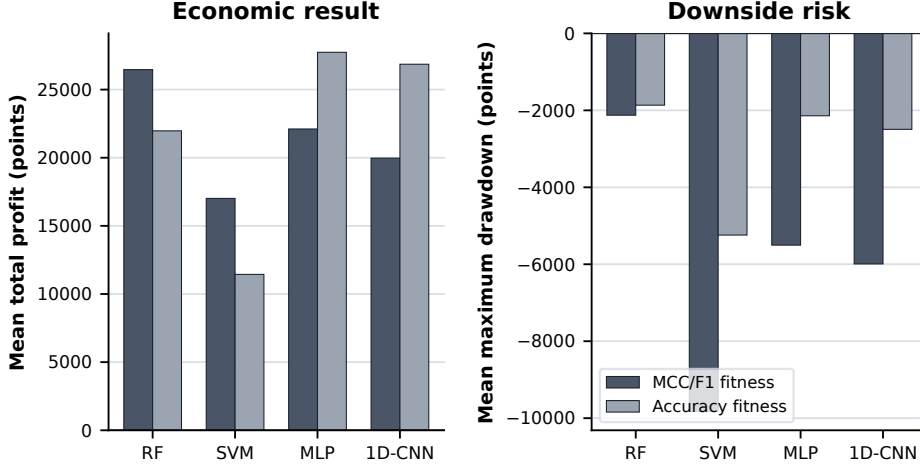


Fig. 12 Mean total profit and maximum drawdown across 15 optimizer-seed blocks per model. Results are reported separately for the two validation objectives.

5 Discussion

This section interprets the results in relation to the methodological sequence established by the IJCNN precursor. The precursor showed that a compact technical-indicator representation and GA-based tuning could produce useful short-term trend predictions. The present benchmark preserves that foundation and tests which conclusions remain valid when the data split, search budget, optimizer family, fitness criterion, and economic endpoint are controlled explicitly.

5.1 Methodological Continuity and Extension

The continuity between the two studies is visible in four stages: the same B3 intraday forecasting problem, construction of short-window technical indicators, Information Gain reduction to seven descriptors, and evolutionary hyperparameter search. The extension is not a replacement of that pipeline, but a stricter experimental comparison. Instead of a randomly sampled training subset and a single GA, the present design retains chronological order, evaluates RF, SVM, MLP, and 1D-CNN, compares RS, GA, PSO, DE, and GWO under 1,000 candidate fits, and keeps the final test block isolated.

5.2 Predictive Performance and Fitness Choice

The descriptive means show a small advantage for the compound MCC/ F_1 protocol in both macro test accuracy (0.623 versus 0.617) and macro test MCC (0.247 versus 0.235). This difference should not be interpreted as universal superiority. The configuration heatmaps contain both positive and negative cells, demonstrating that the selected fitness interacts with the backbone and optimizer. In particular, optimizing validation accuracy does not guarantee higher held-out accuracy, while an imbalance-aware criterion does not dominate every individual configuration.

The within-protocol statistical results provide a more stable conclusion: model identity matters under the controlled blocks. MLP obtains the best mean rank under the accuracy protocol and differs significantly from the other backbones in the exploratory paired comparisons. This finding is consistent with the precursor’s observation that the MLP can represent nonlinear intraday boundaries, but it is now bounded by the common temporal split and evaluation budget.

5.3 Optimizer Search Dynamics

The model-specific convergence figures make the optimizer comparison more informative than a final score alone. They show the search trajectories in the same style as the precursor’s GA-fitness plots while placing every method on the common candidate-evaluation axis. Early improvement, late improvement, and trajectory dispersion can therefore be interpreted as search behaviour rather than as consequences of unequal generation or iteration counts. No optimizer is declared universally best because the relevant trajectory depends on both the backbone and the fitness landscape.

5.4 Economic Meaning of the Predictions

The economic results confirm that predictive and trading rankings are related but not interchangeable. RF obtains the largest mean profit under MCC/ F_1 fitness, whereas MLP obtains the largest mean profit under accuracy fitness. RF nevertheless has the least negative mean drawdown under the latter protocol. Transaction costs and the number of executed signals therefore alter the practical ordering that would be obtained from accuracy alone. A deployment decision must state whether profit, downside risk, turnover, or stability is the primary criterion.

5.5 Limitations and Future Work

The direct dual-fitness contrast is descriptive because only three seeds are common to every model-optimizer cell. The Friedman and Wilcoxon analyses are exploratory model-level tests and do not represent independent market periods. The dataset covers one liquid Brazilian futures contract and one chronological test segment, so transfer to other instruments and regimes remains untested. Finally, the fixed 1,000-evaluation cap establishes budget parity but does not characterize asymptotic behaviour at larger computational budgets.

The next phase should retain the same IJCNN-style sequence and run a preregistered common 20-seed study for every cell. Each candidate evaluation should be

archived with validation MCC, accuracy, F_1 , and economic statistics, followed by paired inference performed separately for each endpoint. This would provide confirmatory evidence while preserving the leakage-free and equal-budget design established here.

6 Conclusion

This study transforms a GA-based conference precursor into a controlled Neural Computing and Applications benchmark for neural intraday trend classification. It preserves the predecessor’s practical insight—a compact, Information-Gain-selected technical representation can support metaheuristic hyperparameter search—but replaces randomized validation and a single-optimizer design with a chronological dual-fitness protocol, five budget-matched optimizers, and locked predictive and economic assessment.

The current evidence indicates that the choice between compound MCC/ F_1 fitness and accuracy fitness produces configuration-dependent changes in both held-out accuracy and MCC. Within the accuracy-fitness protocol, MLP is the strongest model in the exploratory predictive analysis and leads mean total profit, although RF has the smallest mean drawdown. These results reinforce the central methodological lesson: selection objective, reported metric, and economic utility must be treated as related but distinct quantities.

The manuscript deliberately avoids a confirmatory claim about a universal best fitness function because only three paired seeds are currently available across every cell. Completing the common 20-seed design is the necessary next step. The resulting framework nevertheless provides a transparent and reproducible basis for future optimizer comparisons in non-stationary financial time series.

Declarations

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Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

Author Contributions

All authors contributed to the study conception and design. Data collection and experimental execution were performed by [Anonymous for double-blind review]. Statistical analysis and manuscript preparation were carried out by [Anonymous for double-blind review]. All authors read and approved the final manuscript.

References

- [1] Bashir, U., Singh, K., Mansotra, V., Khanday, A.M.U.D., Neshat, M.: EvoBag-Net: a decomposition-driven bagging ensemble with evolutionary hyperparameter optimization for robust stock price prediction. *Neural Computing and Applications* **38**, 31 (2026) <https://doi.org/10.1007/s00521-025-11789-z>
- [2] Jesus Jr., L.C., Fernández-Navarro, F., Carbonero-Ruz, M.: Enhancing financial time series forecasting through topological data analysis. *Neural Computing and Applications* **37**, 6527–6545 (2025) <https://doi.org/10.1007/s00521-024-10787-x>
- [3] Souza, A.V.S., Pinto, R.F., Dalmazo, B.L., Borges, E.N., Lucca, G., Mattos, V.L.D., Berri, R.A.: Predictive analysis with technical indicators and feature selection for futures contracts trading. In: *International Conference on Computational Science and Its Applications (ICCSA)*, pp. 349–367. Springer, ??? (2025)
- [4] Al Bannoud, M., Dias Martins, T., Lima Montalvão, S.A., Annichino-Bizzacchi, J.M., Maciel Filho, R., Wolf Maciel, M.R.: Accelerating hybrid ANN–ODE frameworks using surrogate machine learning and metaheuristic optimization for predicting recurrent venous thromboembolism. *Neural Computing and Applications* **38**, 285 (2026) <https://doi.org/10.1007/s00521-026-12047-6>
- [5] Jarrow, R.A., Oldfield, G.S.: Forward contracts and futures contracts. *Journal of Financial Economics* **9**(4), 373–382 (1981)
- [6] Cardoso, F.C., Malska, J.A.V., Ramiro, P.J., Lucca, G., Borges, E.N., Mattos, V.L.D., Berri, R.A.: Bovdb: a data set of stock prices of all companies in b3 from 1995 to 2020. *Journal of Information and Data Management* **13**(1) (2022)
- [7] Souza, A.V.S., Lucca, G., Borges, E.N., Cardoso, F.C., Dalmazo, B.L., Berri, R.A.: Dataset for Intraday Analysis of B3 Stock Prices. *Harvard Dataverse* (2024). <https://doi.org/10.7910/DVN/TMB4IG>

- [8] Makridakis, S., Spiliotis, E., Assimakopoulos, V.: Statistical and machine learning forecasting methods: Concerns and ways forward. *PLoS ONE* **13**(3), 0194889 (2018)
- [9] Dhingra, S., Gruosso, G., Gajani, G.S.: A benchmark study of optimizers for short-term solar pv power forecasting using neural networks under real-world constraints. *Neural Computing and Applications* **37**, 23909–23939 (2025) <https://doi.org/10.1007/s00521-025-11546-2>
- [10] Billah, M.M., Sultana, A., Bhuiyan, F., Kaosar, M.G.: Stock price prediction: comparison of different moving average techniques using deep learning model. *Neural Computing and Applications* **36**(11), 5861–5871 (2024) <https://doi.org/10.1007/s00521-024-09412-1>
- [11] Box, G.E.P., Jenkins, G.M., Reinsel, G.C., Ljung, G.M.: *Time Series Analysis: Forecasting and Control*. John Wiley & Sons, ??? (2015)
- [12] Jiang, W.: Applications of deep learning in stock market prediction: recent progress. *Expert Systems with Applications* **184**, 115537 (2021)
- [13] Nabipour, M., Nayyeri, P., Jabani, H., Mosavi, A., *et al.*: Predicting stock market trends using machine learning and deep learning algorithms via continuous and binary data: a comparative analysis. *IEEE Access* **8**, 150199–150212 (2020)
- [14] Naik, N., Mohan, B.R.: Optimal feature selection using boruta for stock trend prediction. *Procedia Computer Science* **165**, 688–694 (2019)
- [15] Ecer, F., Ardabili, S., Band, S.S., Mosavi, A.: Training multilayer perceptron with genetic algorithms and particle swarm optimization for modeling stock price index prediction. *Entropy* **22**(11), 1239 (2020) <https://doi.org/10.3390/e22111239>
- [16] Bandgar, S., Deshmukh, R., Kulkarni, P.: Finchronos: Financial time series prediction using multi-model ensemble architectures. *Neural Computing and Applications* **37**(4), 2450–2468 (2025)
- [17] Hunter, J.S.: The exponentially weighted moving average. *Journal of Quality Technology* **18**(4), 203–210 (1986)
- [18] Singh, P., Sharma, R.: Feature engineering and normalization techniques in financial machine learning. *Expert Systems with Applications* **195**, 116548 (2022)
- [19] Lei, Y., Liu, H.: Feature selection for high-dimensional financial data using information gain. *IEEE Transactions on Knowledge and Data Engineering* **24**(3), 432–447 (2012)
- [20] Kumar, V., Minz, S.: Impact of feature selection on metaheuristic hyperparameter tuning. *Applied Soft Computing* **21**, 214–225 (2014)

- [21] Jovanovic, L., Zivkovic, M., Bacanin, N., Dobrojevic, M., Simic, V., Sadasivuni, K.K., Tirkolaee, E.B.: Evaluating the performance of metaheuristic-tuned weight agnostic neural networks for crop yield prediction. *Neural Computing and Applications* **36**, 14727–14756 (2024) <https://doi.org/10.1007/s00521-024-09850-4>
- [22] Faris, H., Hassonah, M.A., Ala'M, M., Mirjalili, S., Al-Zoubi, I.: Grey wolf optimizer: A review of recent variants and applications. *Neural Computing and Applications* **30**(2), 413–435 (2018) <https://doi.org/10.1007/s00521-017-3272-5>
- [23] Rajwar, K., Deep, K., Das, S.: An exhaustive review of the metaheuristic algorithms for search and optimization: taxonomy, applications, and open challenges. *Artificial Intelligence Review* **56**, 13187–13257 (2023) <https://doi.org/10.1007/s10462-023-10470-y>
- [24] Goldberg, D.E.: *Genetic Algorithms in Search, Optimization and Machine Learning*. Addison-Wesley, Reading, MA, USA (1989)
- [25] Storn, R., Price, K.: Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization* **11**(4), 341–359 (1997) <https://doi.org/10.1023/A:1008202821328>
- [26] Kennedy, J., Eberhart, R.: Particle swarm optimization. In: *Proceedings of the IEEE International Conference on Neural Networks*, vol. 4, pp. 1942–1948 (1995). <https://doi.org/10.1109/ICNN.1995.488968>
- [27] Chicco, D., Jurman, G.: The advantages of the matthews correlation coefficient (MCC) over f1 score and accuracy in binary classification evaluation. *BMC Genomics* **21**(1), 6 (2020) <https://doi.org/10.1186/s12864-019-6413-7>
- [28] Henrique, B.M., Sobreiro, V.A., Kimura, H.: Literature review: Machine learning techniques applied to financial market prediction. *Expert Systems with Applications* **124**, 226–251 (2019) <https://doi.org/10.1016/j.eswa.2019.01.012>
- [29] Gad, A.G.: Particle swarm optimization algorithm and its applications: A systematic review. *Archives of Computational Methods in Engineering* **29**(5), 2531–2561 (2022) <https://doi.org/10.1007/s11831-021-09694-4>
- [30] Wang, D., Tan, D., Liu, L.: Particle swarm optimization algorithm: an overview. *Soft Computing* **22**(2), 387–408 (2018) <https://doi.org/10.1007/s00500-016-2474-6>
- [31] Bollinger, J.: Using Bollinger bands. *Stocks & Commodities* **10**(2), 47–51 (1992)
- [32] Mutinda, J., Yong, S.: Hybrid feature selection using stability-enhanced dynamic thresholding for financial market classification. *Journal of Computational Finance* **29**(2), 85–112 (2026)

- [33] Bergstra, J., Bengio, Y.: Random search for hyper-parameter optimization. *Journal of Machine Learning Research* **13**, 281–305 (2012)
- [34] Bareket, O., Pârv, V.: Adaptive machine learning models for market surge detection in dynamic equity environments. *Quantitative Finance* **24**(9), 1415–1432 (2024)
- [35] Rahim, A., Khan, M., Ahmed, S.: Technical indicator parameter optimization using genetic algorithms on high-frequency intraday futures data. *Applied Soft Computing* **152**, 111240 (2025)
- [36] Simatupang, R., Lee, H., Park, S.: Stock market forecasting using bayesian-optimized ensemble decision trees. *Expert Systems with Applications* **268**, 126102 (2026)
- [37] Souza, A.V.S., Dalmazo, B.L., Mattos, V.L.D., Pinto, R.F., Bruno, D.R., Borges, E.N., Lucca, G., Cardoso, F.C., Berri, R.A.: Combining Technical Indicators and Genetic Algorithms for Short-Term Machine Learning Prediction of Mini-Index Futures. Unpublished manuscript (2026)